BMC Immunology

RESEARCH Open Access

Efficacy and safety of gut microbiota-targeted therapy in patients with psoriasis: a systematic review and meta-analysis of randomized controlled trials

Li Zhu[1†], Yanzhen Zhu[2,3†], Jing Xie[1], Yan Zhang[1], Jinmeng Wang[4], Hezi Ji[1] and Xiuyan Wang[5*]

Abstract

Objective Psoriasis is a chronic inflammatory skin condition with significant systemic and psychosocial impacts. Emerging evidence suggests a potential link between gut microbiota and psoriasis, prompting growing interest in gut microbiota-targeted therapies (GMTT) as a potential novel treatment strategy. Aims: This study aims to conduct a systematic review and meta-analysis to evaluate the efficacy and safety of GMTT in patients with psoriasis. Methods: PubMed, Embase, Web of Science, Cochrane, CNKI and Wanfang were searched up to May 2024 for randomized controlled trials (RCTs) assessing GMTT for psoriasis. Outcomes included Psoriasis Area and Severity Index (PASI), Dermatology Life Quality Index (DLQI), C-reactive protein (CRP) and adverse events. The risk of bias of the included RCTs was independently assessed using the Cochrane RoB tool. A random-effects model was used to calculate pooled standardized mean differences (SMDs) and 95% confidence intervals (CIs). Sensitivity, subgroup analyses and publication bias analyses were performed to assess the stability of the results and the sources of heterogeneity. Analyses were conducted using Review Manager 5.4 and STATA 15.1. The GRADE approach was used to assess the certainty of evidence. Results: Eight RCTs involving 409 patients were included. GMTT was associated with a modest reduction in PASI scores (SMD = -0.52, 95% CI [-0.98, -0.06], $I^2 = 79\%$, $p = 0.03$). Subgroup analyses showed greater efficacy with antibiotic intervention, interventions < 3 months, Asian populations and plaque psoriasis. No significant differences were observed in adverse events (RR = 0.99, 95% CI [0.50, 1.94], $I^2 = 23\%$, $p = 0.97$), DLQI or CRP. Funnel plots and Egger's test indicated no significant publication bias, but sensitivity analysis revealed some instability. Conclusions: GMTT may offer modest benefits for psoriasis symptoms, particularly in specific subgroups. However, larger RCTs are needed to confirm these findings due to current limitations such as small sample sizes and regional biases.

Keywords Psoriasis, Gut microbiota-targeted therapy, PASI, RCT, Meta-analysis

†Li Zhu and Yanzhen Zhu contributed equally to this work.

*Correspondence:
Xiuyan Wang
wangxiuyan12@sina.com
[1]Department of Dermatology, The Wenzhou Third Clinical Institute Affiliated To Wenzhou Medical University (Wenzhou People's Hospital), Wenzhou 325000, Zhejiang, China
[2]School of Urban Construction, Hangzhou Polytechnic, Hang Zhou 311402, China
[3]College of Civil Engineering, Zhejiang University of Technology, Hangzhou 310014, China
[4]Department of Dermatology, Wenzhou Central Hospital, Wenzhou 325000, Zhejiang, China
[5]Department of Gastroenterology, The Wenzhou Third Clinical Institute Affiliated To Wenzhou Medical University (Wenzhou People's Hospital), Wenzhou 325000, Zhejiang, China

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Introduction

Psoriasis is a chronic, inflammatory, and immunoregulated condition of the skin and joints, characterized by erythematous, scaly, and pruritic skin plaques [1, 2]. The global prevalence of psoriasis ranges from 0 to 8.5%, varying by age and region [3]. Psoriasis is associated with an elevated risk of comorbidities, including cardiovascular diseases and mental health disorders. Shared inflammatory pathways and genetic susceptibility are considered key mechanisms underlying the association between psoriasis and its comorbidities [4, 5].

Treatment methods for psoriasis mainly include local therapy, phototherapy, oral systemic therapy and biologics systemic therapy. These treatments have substantially improved the quality of life, but there are still many unsolved issues [6].

Recent studies have reported gut microbiota dysbiosis in several inflammatory skin diseases, including atopic dermatitis, rosacea and psoriasis [7]. There is increasing recognition of the gut-skin axis, where intestinal microbes influence skin inflammation and immune modulation through microbial metabolites, systemic immune signaling, and epithelial barrier function. As a result, GMTT, such as probiotics, prebiotics and fecal microbiota transplantation (FMT), have gained attention as potential interventions for psoriasis management [8, 9]. A bidirectional interaction has been noted between the gut and the skin, where gut microbiota affect skin conditions through microbial metabolites, systemic immune signaling, and epithelial barrier function [10, 11]. Therefore, GMTT may be useful for the prevention or treatment of psoriasis.

Gut microbiota diversity is affected by antibiotics, dietary modification, probiotics, prebiotic supplementation, and FMT [12–15]. Jalal Moludi et al. conducted a 2-month randomized controlled trial (RCT) of oral probiotics on 50 patients with psoriasis in 2021. They found that the total Beck's questionnaire and DLQI scores of the probiotic group were significantly improved, and PASI and psoriasis symptom scale (PSS) scores were considerably reduced compared with the placebo group [16]. However, in 2023, Enrico Stefano Suriano et al. assessed the efficacy of adjuvant probiotics in 103 psoriasis patients using probiotics ($n = 50$) and a control group ($n = 53$) and failed to support that probiotics (Lactobacillus rhamnosus) could improve PASI and DLQI in patients with psoriasis ($p = 0.620$) [17]. The above studies all use intestinal-targeted treatments, and all interventions are on the intestinal flora, but their conclusions are inconsistent. A meta-analysis by Zeng et al. in 2024 investigated the efficacy and safety of GMTT in autoimmune and rheumatic diseases. It included only three studies on psoriasis and reported a reduction in PASI scores with probiotics [8]. However, the limited number of psoriasis-specific trials and heterogeneity in interventions limit the generalizability of their findings. Existing studies exhibit substantial heterogeneity in study designs, intervention types, treatment durations and outcome measures, making it difficult to draw consistent conclusions or guide clinical practice.

To date, no comprehensive meta-analysis has synthesized all RCTs exclusively focusing on GMTT for psoriasis. Therefore, we conducted a systematic review and meta-analysis to evaluate the efficacy and safety of GMTT in psoriasis, explore the sources of heterogeneity, and identify potentially responsive subgroups. According to the PICO (Population, Intervention, Comparison, Outcome) framework, this study focused on patients with psoriasis (P) who received GMTTs such as probiotics, antibiotics, or fecal microbiota transplantation (I), and compared with placebo or standard treatments including methotrexate, vitamin D analogs, or biologics (C). The outcomes (O) assessed included PASI, DLQI, CRP, and incidence of adverse events. In addition to evaluating the overall efficacy and safety of GMTT in psoriasis, this study aims to identify optimal patient subgroups and treatment conditions through subgroup analyses. Our findings will help inform future research directions in microbiota-based psoriasis management that may help to help guide clinical decision making.

Materials and methods

Protocol and registration

This systematic review was registered with PROSPERO (CRD42024545661) and conducted under the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, including the extended statement for network meta-analyses [18].

Search methods

Embase, PubMed, Web of Science, Cochrane Library, CNKI and Wanfang databases were searched for RCTs that investigated GMTTs on psoriasis. The search terms included "Probiotics or Synbiotics or Antibiotic or Fecal Transplantation", "Psoriasis", and "Randomized Controlled Trial & Clinical Trial & Intervention Study & Clinical Study", without restrictions on language or region. Additionally, the references of eligible articles were manually searched. The search strategies are displayed in Supplementary Data 1.

Study selection

After the retrieved articles were screened through titles, abstracts, and full texts, eligible articles were selected under the criteria.

Inclusion criteria:

(a) Participants: Individuals aged ≥ 12 years diagnosed with psoriasis through clinical assessment.
(b) Intervention and comparison: control group: conventional treatment of psoriasis (including Methotrexate, Biologics, topical corticosteroids and Vitamin D analogs) or placebo treatment; intervention group: GMTT based on the control therapy.
(c) Reporting at least one outcome indicator: primary outcomes: PASI and adverse event rate; secondary outcomes: CRP and DLQI.

Exclusion criteria:

(a) Non-RCTs, retrospective studies, reviews, and animal assays.
(b) Participants: Non-psoriatic patients.
(c) Intervention: GMTT was also given to the control group.
(d) Outcomes: inaccurate data or incomplete outcome measurement, or the data cannot be obtained from the original authors.
(e) Duplicates.

Data extraction

The retrieved literature was imported into EndNote 20, and data were extracted in Excel. Two researchers (LZ and XY W) independently collected the information on five aspects:

(a) Publication: title, first author, year.
(b) Study features: study design, treatment duration.
(c) Participants: number, age, sex, psoriasis type.
(d) Interventions: drug, dose, frequency, administration route.
(e) Outcomes: primary and secondary indicators. The mean and standard deviation were utilized for continuous variables. The number of events and total cases were utilized for categorical data.

Any disagreements were decided by a third researcher (JX).

Quality assessment

Two researchers (YZ and JM W) independently appraised the quality of included RCTs utilizing the Cochrane Risk of Bias (RoB) tool and cross-checked their results. Seven domains were appraised: random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, reporting bias, and other biases. Each RCT was rated as high risk of bias, low risk of bias, or unclear bias. Any disagreements were tackled by a third researcher (XY W). This three-person assessment process was designed to ensure objectivity, consistency, and transparency.

Statistical analysis

Review Manager 5.4 and Stata 15.1 were leveraged for statistical analyses. The risk ratio (RR) was computed for binary variables, while the mean difference (MD) or standardized mean difference (SMD) was computed for continuous data. Heterogeneity was judged using 95% confidence interval (CI) and I^2 test. A random-effects model was used for all meta-analyses. This model was chosen because of the expected clinical and methodological heterogeneity among the included studies, including differences in patient populations, intervention types (e.g., probiotics, antibiotics, FMT) and treatment durations. The random-effects model allows for more conservative effect estimates, minimizes the risk of false-positive results, and reduces the influence of between-study heterogeneity. A p-value < 0.05 or $I^2 \geq 50\%$ was indicative of significant heterogeneity. Furthermore, sensitivity analysis was implemented via the leave-one-out method. Publication bias was checked via funnel plots and Egger's test. In addition, subgroup analyses were implemented by intervention method, treatment duration, population region, and psoriasis type to explore the stability of the results and the source of heterogeneity. Additionally, according to GRADE, the evidence of each outcome was evaluated and graded as "high", "moderate", "low", or "very low" quality [19].

Results

Research selection

The selection process is manifested in Fig. 1. In total, 1,626 potentially relevant records were screened, including 176 from PubMed, 204 from Embase, 44 from Web of Science, 980 from the Cochrane Library, 80 from CNKI, and 142 from the Wanfang Database. When duplicates were eliminated, the titles and abstracts of 1457 articles were reviewed. Finally, 8 articles involving 409 patients (208 in the intervention group versus 201 in the control group) were enrolled [16, 17]20– [20]. All these articles were RCTs.

Study traits

Patients were from China, India, Iran, Brazil, and Denmark. The treatment duration ranged from 2 to 12 months, and FMT was a one-time procedure. Psoriasis types included plaque psoriasis and arthritis psoriasis. Control interventions encompassed conventional therapy, Actretin Capsules, vitamin C tablets, Methotrexate, Secukinumab, Ustekinumab, Adalimumab, high- and medium-potency corticosteroids and Vitamin D analogs like Calcipotriol. The intervention group received GMTT based on interventions adopted in the control group.

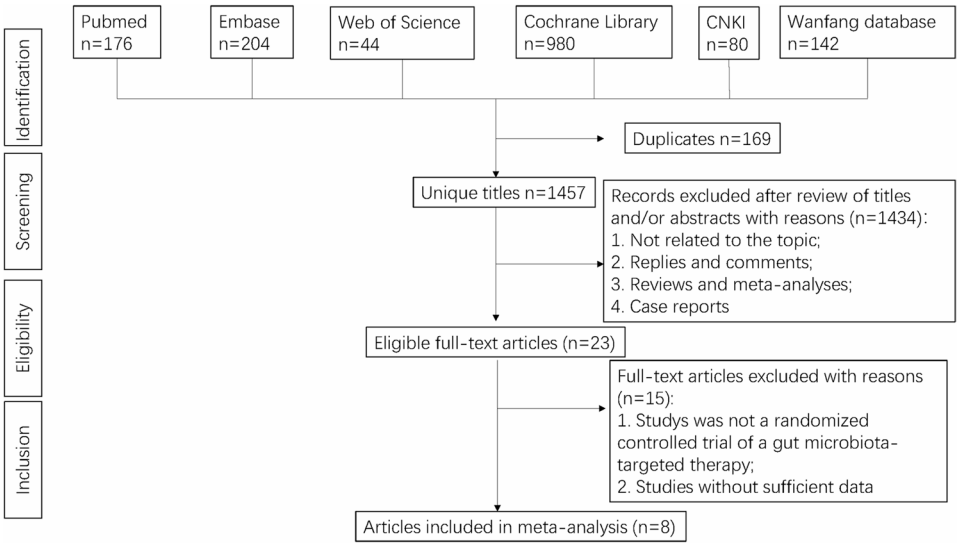


Fig. 1 Flowchart of RCT selection

GMTTs included probiotics, fecal transplants, and antibiotics. Table 1 lists the characteristics of the included RCTs.

Risk of bias

Among the 8 RCTs included, 4 used computer-generated random numbers [16, 17, 23, 25]. The remaining 4 RCTs did not describe the specific method of randomization and had unknown risks. One described the single-blinding method and so had high risk [22]. Other RCTs were double-blinded and rated as low risk. All RCTs had complete outcome data, with a rate of follow-up loss of less than 20%, and thus all had low risk. Three RCTs were not registered [20–22], and it was impossible to determine selective reporting, so they were assessed as unknown risks. Other RCTs were registered and had low risk. No other bias was found in any RCTs, and this domain was rated as low risk (Fig. 2a and b).

Primary outcomes

Change in PASI

All eight RCTs reported changes in PASI scores. PASI scores were decreased after GMTT (SMD = −0.52, 95% CI [−0.98, −0.06], $I^2 = 79\%$, $p = 0.03$) (Fig. 3a). However, the high heterogeneity (I^2 = 79%) indicated notable variability across studies. Subgroup analyses were conducted to explore the potential sources of heterogeneity.

Details of all subgroup analysis results are summarized in Table 2. Subgroup analysis by intervention type indicated that antibiotic therapy (azithromycin) yielded the most significant effect (SMD = −2.42; 95% CI [−3.25, −1.60]; $p < 0.00001$; based on a single study) (Table 2). In contrast, FMT showed no benefit (SMD = −0.03; 95% CI [−0.75, 0.68]; $p = 0.93$) (Table 2).

Subgroup analysis by intervention duration suggested that short-term interventions (< 3 months) were associated with a statistically significant benefit (SMD =−0.36; 95% CI [−0.71, −0.02], $p = 0.04$), while long-term interventions (≥ 3 months) showed no significant effect (Table 2).

Subgroup analysis by region revealed that the effect was significant in Asian populations (SMD = −0.80; 95% CI [−1.47,−0.12]; $I^2 = 84\%$; $p = 0.02$), but not in European populations (Table 2).

Subgroup analysis by psoriasis type suggested that benefits were observed in plaque psoriasis (SMD = −0.59; 95% CI [−1.10, −0.08], $I^2 = 81\%$, $p = 0.02$), but not in psoriatic arthritis (Table 2).

To assess whether differences in control treatment types influenced the efficacy of GMTT, we conducted a subgroup analysis based on the nature of the comparator. Control interventions were categorized as either placebo-only or active comparators (e.g., methotrexate, biologics, topical corticosteroids, and vitamin D analogs). The placebo-only subgroup ($n = 169$) showed a greater reduction in PASI scores (SMD = − 0.78, 95% CI [− 1.69, 0.13], $p = 0.09$, $I^2 = 87\%$), though without statistical significance. The active comparator subgroup ($n = 240$) demonstrated a smaller reduction (SMD = − 0.29, 95% CI [− 0.71, 0.13], $p = 0.17$, $I^2 = 57\%$) (Table 2).

Although subgroup analyses helped identify potential effect modifiers, the limited number of studies in several subgroups, especially for antibiotics and FMT, reduced the reliability of these findings. These results should be interpreted cautiously.

Overall, subgroup analyses suggest that short-term interventions (< 3 months) with probiotics or antibiotics may be more effective in Asian patients with plaque

Table 1 Traits of included RCTs

Article	Sources of patients	Types of Psoriasis	Study type	Age (years)		Sample size		Interventions		Duration (months)	Outcomes
				T	C	T	C	T	C		
Lu et al. 2017 [21]	China	plaque psoriasis	RCT	51.3(28.00)	52.2(29.5)	25	25	Standard-of-care plus probiotics(Bifidobacterium Quadruple Live Tablets, 1 g BID or TID)	Standard-of-care	3	PASI
V.N. Saxena et al. 2010 [22]	India	plaque psoriasis	RCT	13–63*	13–63*	28	15	oral azithromycin(single oral 500 mg daily dose for 4 days, repeat with a gap of 10 days for 48 weeks)	vitamin C tablets	12	PASI
Jalal Moludi et al. 2021 [16]	Iran	plaque psoriasis	RCT	42.70(9.10)	43.10(7.80)	25	25	Probiotic(Lactobacillus acidophilus, Bifidobacterium bifidum, Bifidobacterium lactis and Bifidobacterium langum with 1.8 *10 ^9 CFU, BID for 8 weeks)	malto-dextrin capsules	2	PASI/PSS/CRP/IL-6/TAC/MDA/AE/DLQI/BDI
Jalal Moludi et al. 2022 [23]	Iran	plaque psoriasis	RCT	42.04 (8.10)	43.76(8.86)	23	23	Probiotic(multi-strain at least 1.6 × 109 CFU/g of probiotics, including (Lactobacillus acidophilus, Bifidobacterium bifidum, Bifidobacterium lactis, and Bifidobacterium langum) for eight weeks used BID)	malto-dextrin capsules	2	PASI/CRP/FBS/TC/TG/LDL/HDL/DBP/SBP/IL1-Beta/LPS
Ali Akbarzadeh et al. 2022 [24]	Iranian	plaque psoriasis	RCT	44.16(2.18)	38.25(1.79)	25	27	Lactocare® probiotic	placebo	3	PASI/VAS/DLQI
Gilli et al. 2023 [20]	Brazil	plaque psoriasis	RCT	51.67(14.88)	53.29(14.59)	18	17	Standard-of-care plus probiotic(5 billion CFU/g of L rhamnosus Lr-G14 1x/day for 60 days)	Standard-of-care + placebo	2	PASI/BSA/DLQI/IL-23/IL-17

Table 1 (continued)

Article	Sources of patients	Types of Psoriasis	Study type	Age (years)		Sample size		Interventions		Duration (months)	Outcomes
				T	C	T	C	T	C		
Enrico Stefano Suriano et al. 2023 [17]	Brazil		RCT	49.6(12.48)	51.24(13.02)	50	53	standard-of-care plus probiotics(Lactobacillus rhamnosus 3*10^6 per day)	standard-of-care plus placebo	6	PASI/DLQI
Kragsnaes et al. 2024 [25]	Danish	peripheral PsA	RCT	48.9(16.1)	52.4(11.0)	14	16	one gastroscopic-guided FMT	one gastroscopic-guided sham transplantation	One time	PASI/CRP/DLQI/Treatment failure/Total patients receiving > 1 IA glucocorticoid/Total patients starting non-methotrexate conventional synthetic DMARD(s) injection/HAQ-DI/ACR20/50/70 response/SPARCC/Digits affected with dactylitis/PsARC/VAS

Data are mean (SD), *ages range

Psoriasis Area and Severity Index, *PSS* Psoriasis Symptom Scale, *TAC* Total Antioxidant Capacity, *MDA* malon- dialdehyde, *AE* Adverse Events, *DLQI* Dermatology Life Quality Index, *BDI* Beck's questionnaire, *FBS* Fasting Blood Glucose, *TC* Total Cholesterol, *TG* Triglyceride, *LDL* Low Density Lipoprotein, *HDL* How Density Lipoprotein, *DBP* Diastolic Blood Pressure, *SBP* Systolic Blood Pressure, *LPS* Lipopolysaccharides, *VAS* Visual Analogue Scale, *BSA* Body Surface Area, *PsA* Psoriatic Arthritis, *FMT* Faecal Microbiota Transplantation, *DMARD* Disease-Modifying Antirheumatic Drug, *HAQ-DI* Health Assessment Questionnaire Disability Index, ACR20/50/70 response, ≥20%/50%/70% improvement from baseline in the number of tender and swollen joints and in at least 3 of 5 other specified domains, *SPARCC* Spondyloarthritis Research Consortium of Canada, *PsARC* Psoriatic Arthritis Response Criteria, *CFU* colony forming units, *BID* twice a day, *TID* 3 times per day, Lactocare® capsules containing 1×10^9 CFU of Lactobacillus casei, Lactobacillus acidophilus, Lactobacillus rhamnosus, Lactobacillus bulgaricus, Bifidobacterium breve, Bifidobacterium longum, Streptococcus thermophiles

psoriasis. However, the limited number of studies in each subgroup prevents drawing definitive conclusions.

Adverse events

3 RCTs reported the incidence of adverse events. No statistical difference was noted between the intervention and control groups (RR = 0.99, 95% CI [0.50, 1.94], $I^2 = 23\%$, $p = 0.97$) (Fig. 3b). No serious adverse events were reported [23, 24].

Secondary outcomes

Change in DLQI

5 RCTs analyzed DLQI. No considerable difference was noted in improving patients' quality of life (SMD = −0.26, 95% CI [−0.73, 0.21], $I^2 = 69\%$, $p = 0.27$) (Fig. 3c). This result should be interpreted with caution due to moderate heterogeneity and insufficient statistical power. Furthermore, the clinical relevance of this small effect size remains unclear.

Change in CRP

3 RCTs analyzed CRP. No considerable difference was noted in reducing CRP (SMD = −0.35, 95% CI [−1.17, 0.47], $I^2 = 80\%$, $p = 0.40$) (Fig. 3d). Given the limited number of studies and high heterogeneity, this result should be interpreted as inconclusive rather than negative.

Publication bias and sensitivity analysis

Funnel plots were symmetric for PASI, DLQI, and CRP (Fig. 4a and c), denoting no publication bias. Egger's test indicated no significant publication bias for PASI ($p = 0.073$), DLQI ($p = 0.247$), and CRP ($p = 0.093$) (Fig. 5a and c). Sensitivity analysis found that the sensitivity came from two articles [23, 24]. If Ali Akbarzadeh 2022 (PASI,

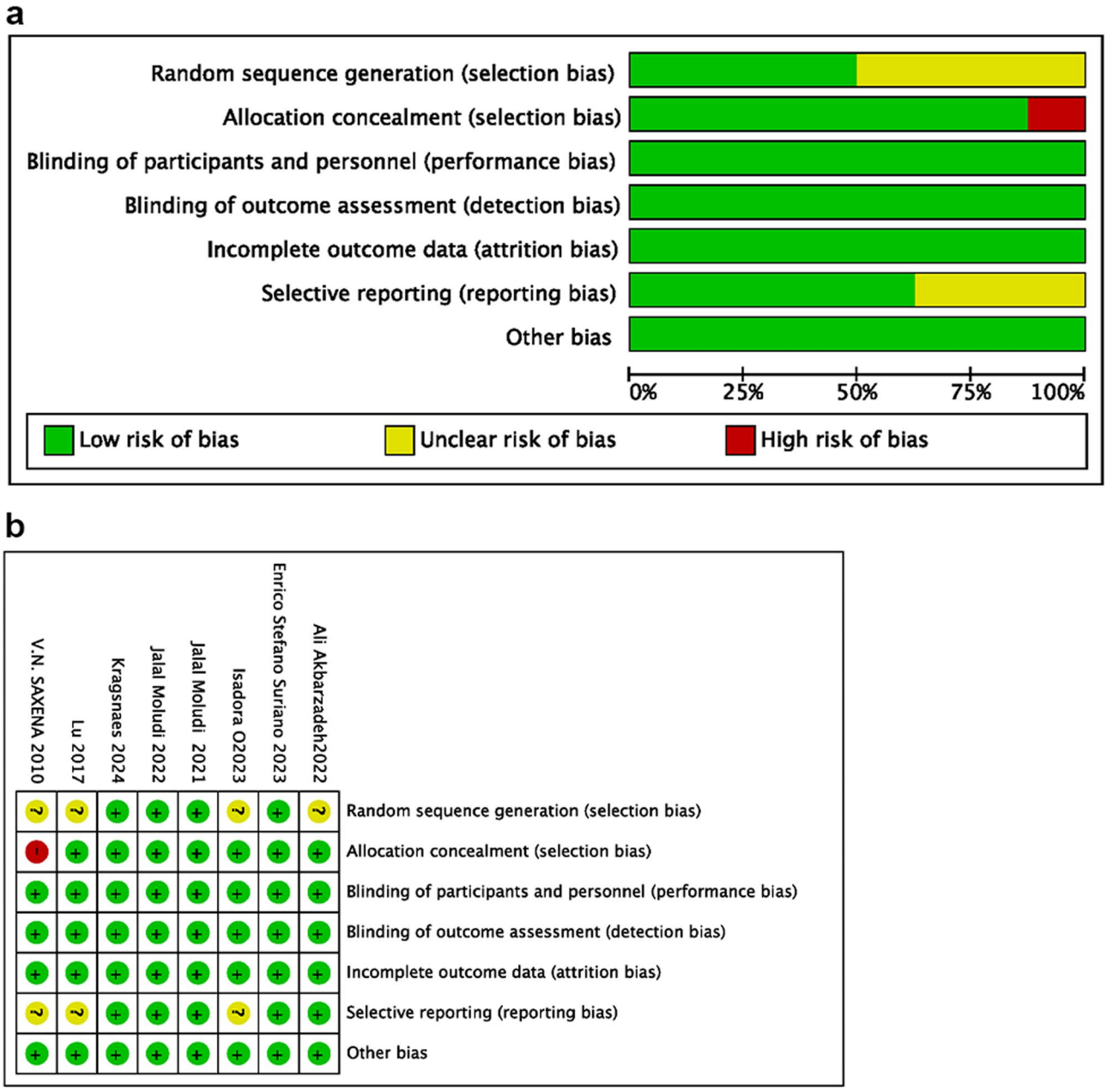


Fig. 2 Cochrane risk of bias assessment of RCT

SMD = −0.47, 95% CI [−0.97, 0.04]) and Jalal Moludi 2022 (PASI, SMD = −0.52, 95% CI [−1.05, 0.01]) were excluded, the significant results became insignificant (Fig. 6).

GRADE rating

The certainty of evidence for each outcome was evaluated using the GRADE framework. The quality of evidence was rated as low for PASI, DLQI and CRP, primarily due to serious inconsistency and imprecision. In contrast, the evidence for adverse events was rated as moderate, indicating a relatively high level of confidence in the safety findings (Table 3).

Discussion

Psoriasis is a chronic, non-contagious, pruritic, and painful disease with no cure and multiple complications. Although microtrauma, sunburn, infection, systemic medications, and stress can trigger psoriasis, genetic predisposition and immunity are critical in the onset of psoriasis [6, 26]. A 12-week RCT conducted by Vicente Navarro-López on 90 patients with psoriasis found that 66.7% of patients receiving probiotics reached a 75%

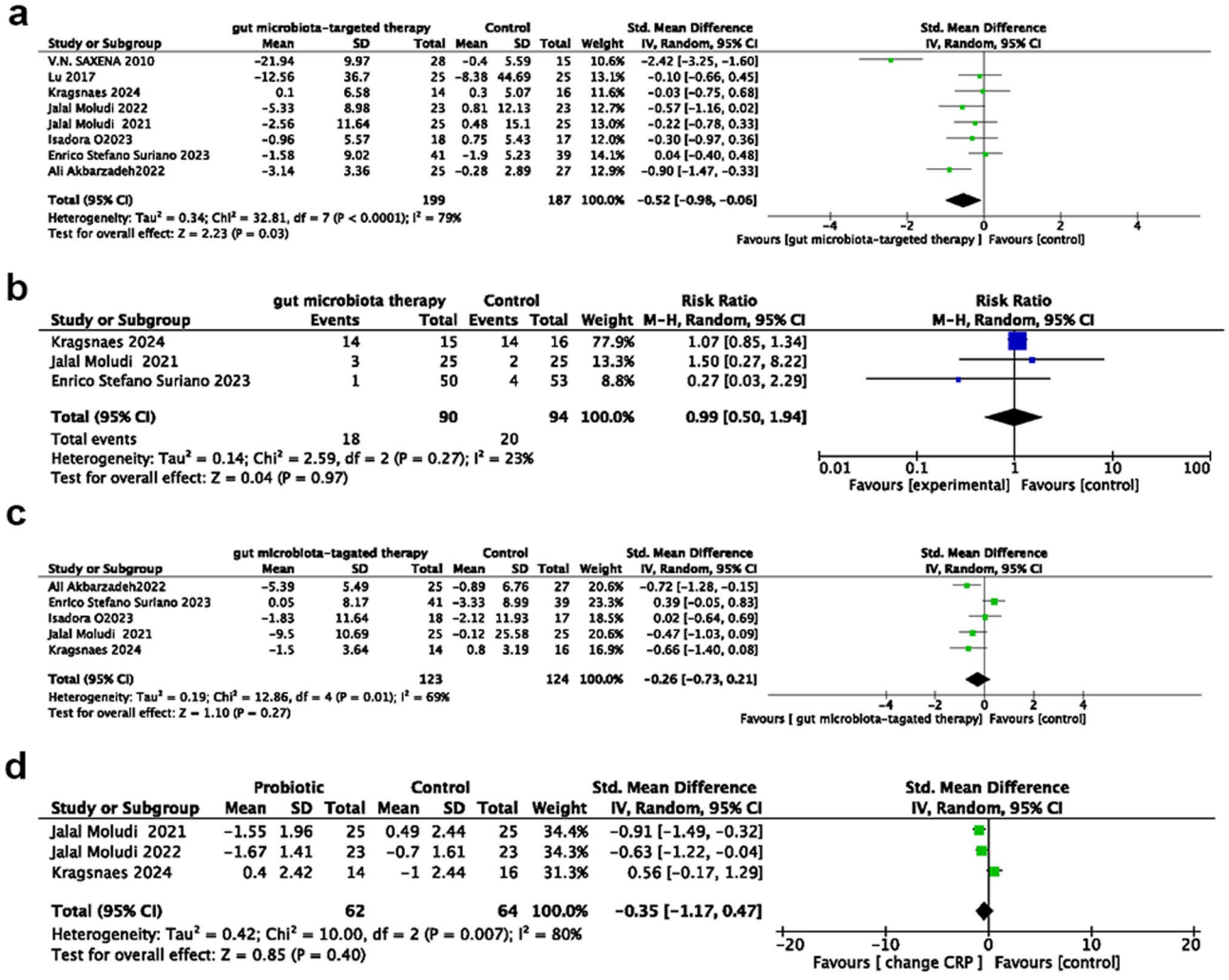


Fig. 3 The results of PASI, adverse events, DLQI, CRP

decrease in PASI scores, while only 41.9% in the placebo group ($p < 0.05$). Additionally, notable shifts in gut microbiota compositions were observed: Micromonospora and Rhodococcus disappeared, while Collinsella and Lactobacillus—genera associated with intestinal health—were enriched post-treatment [27]. Hence, this review aimed to investigate whether GMTTs, including probiotics, are beneficial for patients with psoriasis.

Our meta-analysis demonstrated a moderate reduction in PASI scores (SMD = − 0.52) after interventions with GMTTs. Egger's test did not exhibit any pronounced publication bias. While this suggests a potential clinical benefit, especially considering that a 5–10 point reduction in PASI is generally regarded as meaningful in moderate-to-severe psoriasis, the interpretation should be cautious due to limited robustness. In six of the eight included RCTs, PASI reductions were significant, with several studies showing absolute decreases greater than 5 points. Our findings align with a previous meta-analysis by Wei et al. [9], which included five probiotic studies and reported improvements in both PASI and DLQI. By incorporating additional Chinese studies and expanding the dataset to 211 patients, we sought to improve the comprehensiveness of the evidence. However, our results were not consistent across all subgroups or intervention types.

In contrast to PASI, DLQI and CRP were not significantly improved after interventions with GMTTs. Several factors may explain these null results: (1) most interventions lasted no longer than 12 weeks, potentially insufficient to influence systemic inflammation or patient-perceived quality of life; (2) the limited number of studies assessing DLQI and CRP weakened statistical power [28]; and (3) PASI improvements do not necessarily correlate with changes in systemic markers or subjective well-being. Moreover, CRP is a nonspecific inflammatory biomarker and may not fully reflect local or immune-modulatory changes induced by GMTTs [5, 28].

Table 2 Subgroup analysis of PASI score for psoriasis

Subgroup	Change in PASI score			
	Study	SMD [95%CI]	*P* value	I^2
Total	8(*n* = 409)	−0.52 [−0.98,−0.06]	0.03	79%
intervention				
probiotics	6(*n* = 336)	−0.32 [−0.60, −0.03]	0.03	37%
azithromycin	1(*n* = 43)	−2.42 [−3.25, −1.60]	< 0.00001	NA
FMT	1(*n* = 30)	−0.03[−0.75, 0.68]	0.93	NA
Therapeutic time				
≥ 3 months	4(*n* = 278)	−0.80 [−1.72, 0.12]	0.09	90%
<3 months	3(*n* = 131)	−0.36[−0.71, −0.02]	0.04	0
Region				
Asia	5(*n* = 241)	−0.80 [−1.47>, −0.12]	0.02	84%
Europe	1(*n* = 30)	−0.03[−0.75, 0.68]	0.93	NA
America	2(*n* = 138)	−0.06 [−0.43>, 0.30]	0.74	0
Population				
plaque psoriasis	7(*n* = 379)	−0.59 [−1.10, −0.08]	0.02	81%
peripheral PsA	1(*n* = 30)	−0.03 [−0.75, 0.68]	0.93	NA
Control intervention				
active comparator	4(*n* = 240)	−0.29 [−0.71, 0.13]	0.17	57%
placebo-only	4(*n* = 169)	−0.78 [−1.69, 0.13]	0.09	87%

PASI psoriasis area severity index, *SMD* Standard Mean Difference, *CI* confidence interval, *FMT* Faecal microbiota transplantation, *PsA* psoriatic arthritis

Our subgroup analyses (Table 2) uncovered that the effect of GMTT (in terms of PASI) was ineffective in the subgroups of FMT, intervention time ≥ 3 months, the European and American populations, and psoriatic arthritis, but the effect was significant in other subgroups. The total I^2 of the article was 79%. These findings should therefore be interpreted cautiously. Importantly, the observed efficacy in some subgroups—particularly the antibiotic intervention group—was derived from a single study, limiting the generalizability of this finding. Additionally, wide confidence intervals in several subgroups further constrain the ability to draw firm clinical conclusions. These limitations underscore the need for larger, well-designed RCTs to verify potential subgroup effects and identify which patient populations may benefit most from GMTTs. Sensitivity analyses revealed that the statistical significance of PASI reduction was largely dependent on two studies [23, 24], and the exclusion of either study rendered the pooled effect non-significant. This fragility of the overall conclusion highlights the need for more robust evidence.

Furthermore, the included studies varied in their designs, control treatments, and GMTT formulations, which may have introduced additional variability in effect sizes. The heterogeneity observed is likely multifactorial. Differences in baseline characteristics, intervention protocols (e.g., strain-specific probiotic differences, dosing regimens), outcome measurements, and regional dietary patterns may contribute to this variability. The diversity of control interventions, including biologics, topical agents, and placebo, further complicates the interpretation of relative effectiveness. Although the Cochrane RoB tool indicated generally low to moderate risk, several RCTs had unclear allocation concealment, insufficient blinding details, and a lack of prospective registration, raising concerns about potential selective reporting bias and limited transparency.

Animal experiments have shown that probiotic supplementation can make the skin healthier by up-regulating the anti-inflammatory factor IL-10, which is known to down-regulate the pro-inflammatory factor IL-17 [29]. IL-17 levels are significantly elevated in skin lesions of patients with psoriasis. It is primarily secreted by T helper 17 cells and can activate keratinocytes and promote their excessive proliferation [30]. Biologically, the gut-skin axis plays a pivotal role in the pathogenesis of psoriasis, primarily through immune modulation and metabolic signaling pathways [31, 32]. GMTTs may exert therapeutic effects by modulating the balance between pro-inflammatory and anti-inflammatory cytokines, notably IL-17 and IL-10, as well as by regulating T-cell activation and maintaining intestinal barrier integrity [33]. Emerging evidence indicates that specific probiotics can enhance regulatory T-cell activity, increase microbial diversity, and attenuate systemic inflammation [34]. However, these mechanisms remain to be conclusively

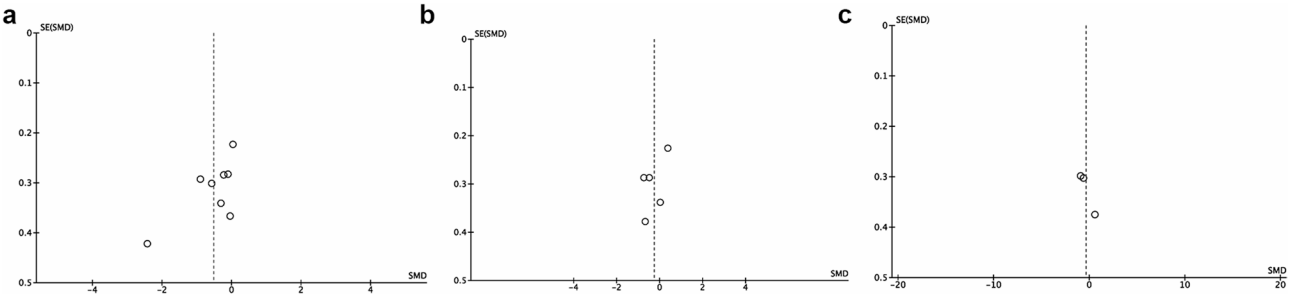


Fig. 4 Funnel plot

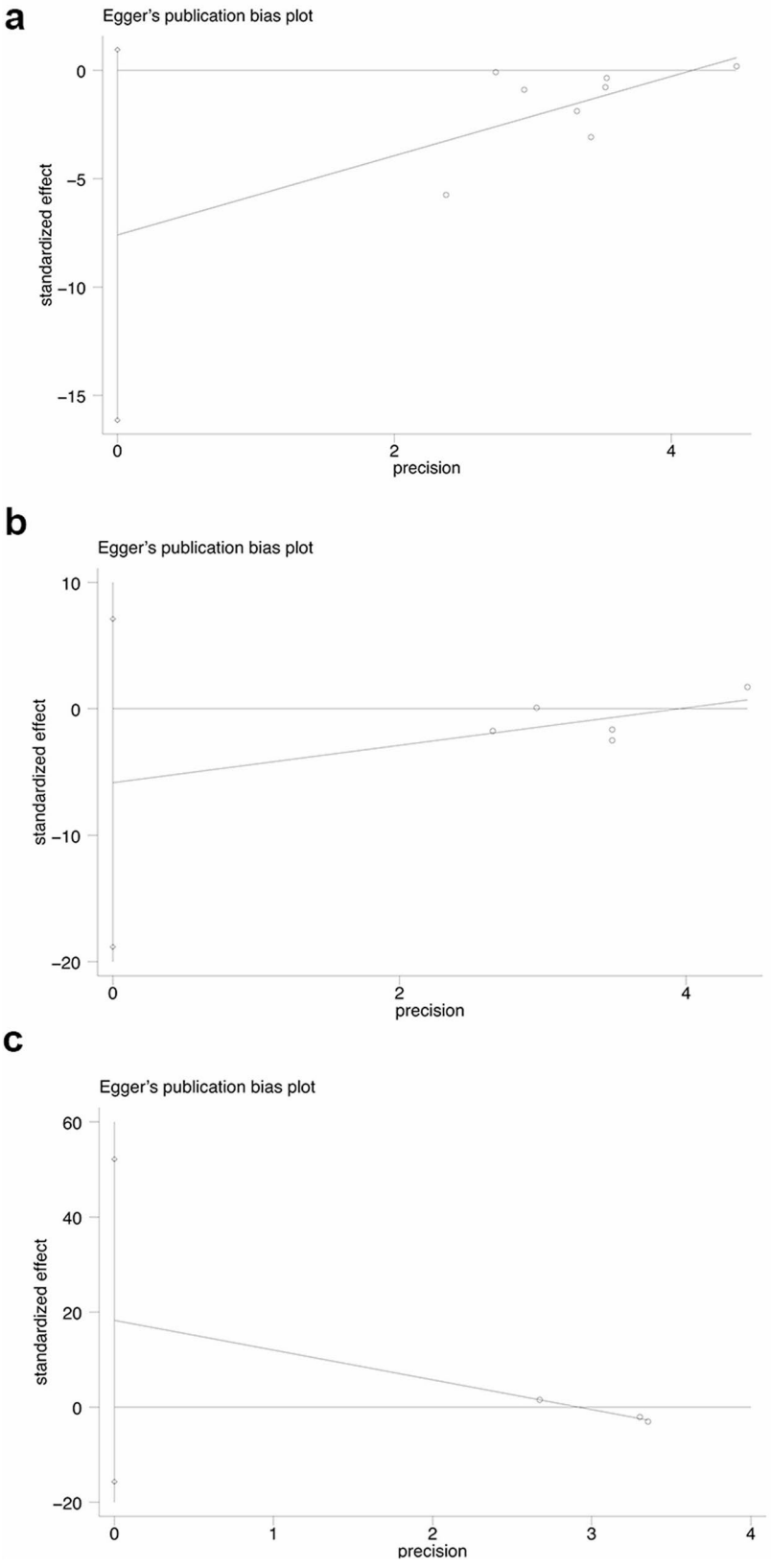


Fig. 5 Egger plot

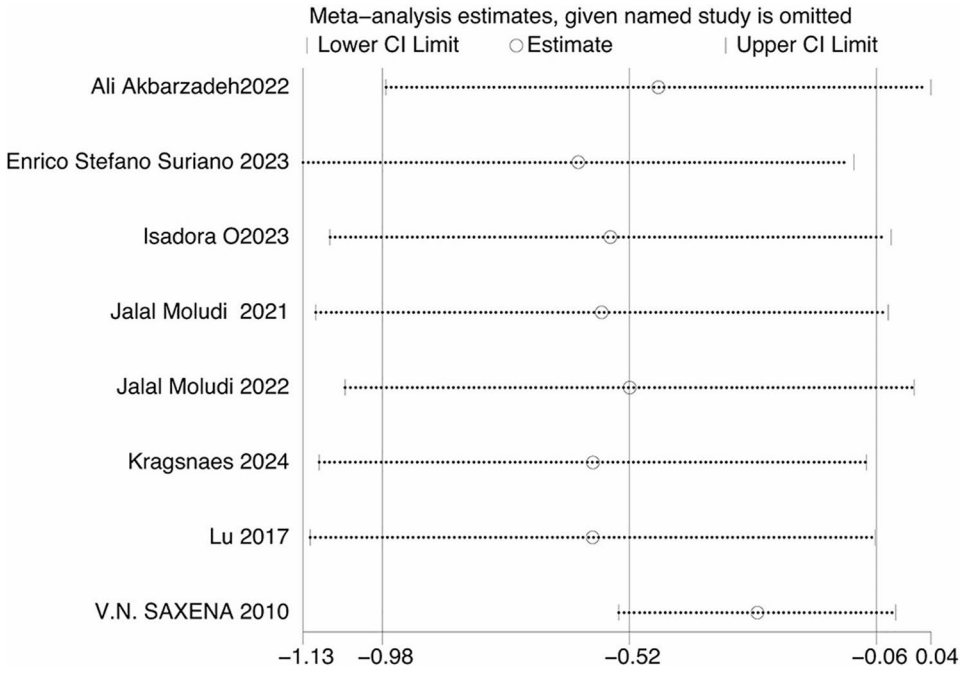


Fig. 6 Sensitivity analysis of PASI

established, as consistent supporting data from microbiome sequencing or immunological profiling are lacking in current research [34]. The biologic agent secukinumab, which blocks IL-17, significantly improves symptoms of psoriasis [35]. Other treatments for psoriasis, such as traditional medicines and phototherapy, have also significantly improved microbial diversity and abundance, which approximates the microbiota of healthy individuals [36]. Our analysis also revealed that probiotics improved the clinical manifestations of psoriasis, while gastroscopy-guided FMT, which also regulates diverse intestinal flora, did not improve the PASI scores and ACR20 response (≥ 20%/improvement from baseline in the number of tender and swollen joints and at least 3 of 5 other specified domains) in patients with psoriatic arthritis. This may be related to donor-recipient matching of FMT and the irritation of the gastrointestinal mucosa during gastroscopy, which affects the gastrointestinal microenvironment [37].

Although our analysis showed that probiotic supplementation < 3 months improved PASI in patients with psoriasis, probiotic supplementation for over 3 months was ineffective. Research shows that short-term supplementation of probiotics may change the colonization pattern of the intestinal mucosa, while long-term supplementation of probiotics cannot sustainably affect the intestinal mucosa and may change intestinal microbial diversity [38]. Therefore, the course of probiotic supplementation for psoriasis should be controlled within 3 months as much as possible. Interestingly, the larger effect size in the < 3 months group (SMD = −0.36) may partially reflect the over-representation of studies conducted in Asian populations, which showed greater PASI reductions in the regional subgroup analysis. This suggests that the apparent superiority of short-term interventions may not be solely due to treatment duration but potentially confounded by geographic or ethnic factors such as diet, genetic predisposition, or baseline microbiota composition. However, due to the small number of studies per subgroup, interaction effects between treatment duration and region could not be assessed.

Furthermore, while GMTT was categorized into probiotics, antibiotics, and fecal FMT, each of these interventions has distinct mechanisms of action. Probiotics function primarily by modulating immune responses, enhancing gut epithelial barrier integrity, and producing anti-inflammatory metabolites. Specific probiotic strains can interact directly with dendritic cells, regulatory T cells, and macrophages, thus suppressing pro-inflammatory cytokines such as IL-17 and TNF-α, which are highly relevant in the pathogenesis of psoriasis [39]. Antibiotics, such as azithromycin in the included studies, transiently reduce microbial diversity and suppress potentially pro-inflammatory cytokines, which may indirectly benefit psoriasis through microbiome reshaping [15]. FMT, by transferring an entire donor microbiota ecosystem, aims for global restoration of gut microbial balance but is subject to variability in donor-recipient compatibility and delivery protocols [40].

According to the subgroup analysis of different regions, GMTT was significantly better for Asian psoriasis patients than for European and American populations.

Table 3 GRADE rating of each outcome

No. of studies	Outcomes	SMD/RR	95%CI	I^2; *P* value	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Plausible confounding	Magnitude of effect	Dose-response gradient	GRADE
8	PASI	−0.52	−0.98, −0.06	79%; *P* < 0.0001	No serious risk	Serious inconsistency	No serious indirectness	Serious imprecision	Undetected	Would not reduce effect	No	No	Low
3	Adverse events	0.99	0.50, 1.94	23%, *P* = 0.27	No serious risk	No serious inconsistency	No serious indirectness	Serious imprecision	NA	Would not reduce effect	No	No	Moderate
5	DLQI	−0.26	−0.73, 0.21	69%, *P* = 0.01	No serious risk	Seriousinconsistency	No seriousindirectness	Seriousimprecision	Undetected	Would notreduce effect	No	No	Low
3	CRP	−0.35	−1.17,0.47	80%; *P* = 0.07	No serious risk	Serious inconsistency	No serious indirectness	Serious imprecision	Undetected	Would not reduce effect	No	No	Low

This may be related to differences in the genetic backgrounds and incidence in different populations. The incidence of psoriasis is higher in high-income countries, e.g., 0.1% in East Asia and 1.5% in Western Europe [6]. The HLA-Cw6 allele is most likely to correlate with psoriasis susceptibility and is more frequent in Caucasian patients than in Asian populations. The HLA-Cw6 allele is most likely to correlate with psoriasis susceptibility and is more frequent in Caucasian patients than in Asian populations [41]. While this difference is well established, it remains unclear whether HLA-Cw6 status influences the responsiveness to GMTT. It is plausible that host genetic factors, including HLA-Cw6, may shape microbiota composition or immune reactivity to microbial modulation [42]. However, none of the included studies stratified outcomes by genotype, and no direct evidence linked HLA-Cw6 expression to GMTT efficacy. Future studies incorporating host genetic profiling alongside microbiota analysis could help clarify this potentially important interaction.

Of course, our article still has certain limitations. The overall sample size is relatively small, and while all included studies were RCTs, many were underpowered, had short follow-up periods and lacked standardization in intervention protocols and outcome assessments. Most intervention methods are probiotics, and relatively few articles focus on FMT and antibiotics. Moreover, the geographic concentration of studies—primarily in Asian countries—raises concerns about generalizability, as cultural dietary habits, baseline microbiota, and genetic backgrounds may influence treatment response. This geographic bias, along with the scarcity of data from other regions, underscores the need for more diverse and methodologically rigorous research across different populations.

The variability in control interventions was further examined through subgroup analysis. Interestingly, a larger effect size was observed in the placebo-only subgroup compared to the active comparator subgroup, although neither result reached statistical significance. This suggests that the perceived efficacy of GMTT may, in part, depend on the strength or nature of the comparator. The greater heterogeneity in the placebo group ($I^2 = 87\%$) may reflect differences in study design, patient populations, or intervention adherence. These findings emphasize the need for future RCTs to adopt more standardized control arms to minimize confounding and allow for more reliable comparisons across trials.

These findings suggest that GMTT may offer modest benefits in selected patient populations. Meanwhile, due to the limitations discussed especially high heterogeneity, reliance on a few influential studies, limited study quality, and regional concentration, the overall evidence should be interpreted with caution. Future large-scale,

multicenter trials with standardized protocols and extended follow-up durations are needed to strengthen the robustness and generalizability of these conclusions.

Conclusion

In conclusion, GMTTs could potentially offer modest clinical benefits, particularly in selected patient populations, though current evidence remains insufficient for definitive recommendations. However, the observed effects are limited by substantial heterogeneity, regional bias, and sensitivity to specific studies. The current evidence is insufficient to draw definitive conclusions.

To enhance the robustness and generalizability of our findings, future research should focus on large-scale, multicenter RCTs with standardized GMTT protocols, longer follow-up periods, and uniform outcome measures. Mechanistic studies incorporating microbiome sequencing and immunologic profiling are also needed to elucidate the biological pathways linking gut microbiota to psoriasis pathogenesis. Until then, GMTTs should be considered exploratory and individualized within clinical decision-making.

From a clinical perspective, these results suggest that GMTT could be considered as an adjunctive therapy in patients with moderate psoriasis who show limited response to conventional treatments. Nevertheless, its application should remain within research settings until more robust trials become available.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12865-025-00747-y.

Supplementary Material 1

Authors' contributions

Xiuyan Wang: Conceptualization, Methodology, Software. Yan Zhang: Validation. Jinmeng Wang: Investigation. Jing Xie: Data Curation, Hezi Ji: Resources. Li Zhu: Writing - Original Draft, Writing - Review & Editing, Funding acquisition. Yanzhen Zhu: Formal analysis, Supervision, Visualization, Project administration.

Funding

This research was supported by the Wenzhou Science and Technology Bureau (Project No.: Y2023463).

Data availability

All data generated or analysed during this study are included in this published article and its supplementary information files.

Declarations

Ethics approval and Consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 22 April 2025 / Accepted: 25 July 2025
Published online: 26 September 2025

References

1. Langley RG, Krueger GG, Griffiths CE. Psoriasis: epidemiology, clinical features, and quality of life. Ann Rheum Dis. 2005;64 Suppl 2(Suppl 2):ii18-23-discussion ii4-5. https://doi.org/10.1136/ard.2004.033217.
2. Korman NJ. Management of psoriasis as a systemic disease: what is the evidence? Br J Dermatol. 2020;182(4):840–8. https://doi.org/10.1111/bjd.18245.
3. Parisi R, Symmons DP, Griffiths CE, Ashcroft DM, Identification P, et al. Global epidemiology of psoriasis: a systematic review of incidence and prevalence. J Invest Dermatol. 2013;133(2):377–85. https://doi.org/10.1038/jid.2012.339.
4. Bu J, Ding R, Zhou L, Chen X, Shen E. Epidemiology of psoriasis and comorbid diseases: a narrative review. Front Immunol. 2022;13:880201. https://doi.org/10.3389/fimmu.2022.880201.
5. Takeshita J, Grewal S, Langan SM, Mehta NN, Ogdie A, Van Voorhees AS, et al. Psoriasis and comorbid diseases: epidemiology. J Am Acad Dermatol. 2017;76(3):377–90. https://doi.org/10.1016/j.jaad.2016.07.064.
6. Griffiths CEM, Armstrong AW, Gudjonsson JE, Barker J. Psoriasis. Lancet. 2021;397(10281):1301–15. https://doi.org/10.1016/S0140-6736(20)32549-6.
7. Thye AY, Bah YR, Law JW, Tan LT, He YW, Wong SH, et al. Gut-skin axis: unravelling the connection between the gut microbiome and psoriasis. Biomedicines. 2022. https://doi.org/10.3390/biomedicines10051037.
8. Zeng L, Yang K, He Q, Zhu X, Long Z, Wu Y, et al. Efficacy and safety of gut microbiota-based therapies in autoimmune and rheumatic diseases: a systematic review and meta-analysis of 80 randomized controlled trials. BMC Med. 2024;22(1):110. https://doi.org/10.1186/s12916-024-03303-4.
9. Wei K, Liao X, Yang T, He X, Yang D, Lai L, et al. Efficacy of probiotic supplementation in the treatment of psoriasis-A systematic review and meta-analysis. J Cosmet Dermatol. 2024;23(7):2361–7. https://doi.org/10.1111/jocd.16299.
10. Salem I, Ramser A, Isham N, Ghannoum MA. The gut microbiome as a major regulator of the gut-skin axis. Front Microbiol. 2018;9:1459. https://doi.org/10.3389/fmicb.2018.01459.
11. O'Neill CA, Monteleone G, McLaughlin JT, Paus R. The gut-skin axis in health and disease: A paradigm with therapeutic implications. BioEssays. 2016;38(11):1167–76. https://doi.org/10.1002/bies.201600008.
12. Sonnenburg JL, Bäckhed F. Diet-microbiota interactions as moderators of human metabolism. Nature. 2016;535(7610):56–64. https://doi.org/10.1038/nature18846.
13. Suez J, Zmora N, Zilberman-Schapira G, Mor U, Dori-Bachash M, Bashiardes S, et al. Post-Antibiotic gut mucosal Microbiome reconstitution is impaired by probiotics and improved by autologous FMT. Cell. 2018;174(6):1406–e2316. https://doi.org/10.1016/j.cell.2018.08.047.
14. Davani-Davari D, Negahdaripour M, Karimzadeh I, Seifan M, Mohkam M, Masoumi SJ, et al. Prebiotics: definition, types, sources, mechanisms, and clinical applications. Foods. 2019. https://doi.org/10.3390/foods8030092.
15. Fishbein SRS, Mahmud B, Dantas G. Antibiotic perturbations to the gut microbiome. Nat Rev Microbiol. 2023;21(12):772. https://doi.org/10.1038/s41579-023-00933-y.
16. Moludi J, Khedmatgozar H, Saiedi S, Razmi H, Alizadeh M, Ebrahimi B. Probiotic supplementation improves clinical outcomes and quality of life indicators in patients with plaque psoriasis: a randomized double-blind clinical trial. Clin Nutr ESPEN. 2021;46:33–9. https://doi.org/10.1016/j.clnesp.2021.09.004.
17. Suriano ES, Souza MDM, Kobata CM, Santos FHY, Mimica MJ. Efficacy of an adjuvant *Lactobacillus rhamnosus* formula in improving skin lesions as assessed by PASI in patients with plaque psoriasis from a university-affiliated, tertiary-referral hospital in Sao Paulo (Brazil): a parallel, double-blind, randomized clinical trial. Arch Dermatol Res. 2023;315(6):1621–9.
18. Brian Hutton P, MSc G, Salanti PD, Caldwell DM, PhD MA, Anna Chaimani BA, Schmid PDCH, Cameron PDC, MSc, John PA, Ioannidis MD, DSc S, Straus MD, MSc K, Thorlund PD, Jeroen P, PharmD, Peter C, Gøtzsche MD, MSc K, Dickersin G. The PRISMA Extension Statement for Reporting of Systematic Reviews Incorporating Network Meta-analyses of Health Care Interventions: Checklist and Explanations. Annals of Internal Medicine. 2015;162(11):777-84. https://doi.org/10.7326/m14.2385%m26030634.

19. Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. introduction-grade evidence profiles and summary of findings tables. J Clin Epidemiol. 2011;64(4):383–94. https://doi.org/10.1016/j.jclinepi.2010.04.026.
20. Gilli IO, da Silva GC, Mendes V, Duarte MG, Tanaka AA. The role of probiotics as an adjunctive therapy in psoriasis. J Psoriasis Psoriatic Arthritis. 2023;8(2):49–55.
21. Saxena VN, Dogra J. Long-term oral azithromycin in chronic plaque psoriasis: a controlled trial. Eur J Dermatol. 2010;20(3):329–33. https://doi.org/10.1684/ejd.2010.0930.
22. Moludi J, Fathollahi P, Khedmatgozar H, Tabrizi FPF, Zare AG, Razmi H, et al. Probiotics supplementation improves quality of life, clinical symptoms, and inflammatory status in patients with psoriasis. J Drugs Dermatol. 2022;21(6):637–44.
23. Kragsnaes MS, Kjeldsen J, Horn HC, Munk HL, Pedersen JK, Just SA, et al. Safety and efficacy of faecal microbiota transplantation for active peripheral psoriatic arthritis: an exploratory randomised placebo-controlled trial. Ann Rheum Dis. 2021;80(9):1158–67.
24. Akbarzadeh A, Alirezaei P, Doosti-Irani A, Mehrpooya M, Nouri F. The efficacy of Lactocare® synbiotic on the clinical symptoms in patients with psoriasis: A randomized, Double-Blind, Placebo-Controlled clinical trial. Dermatol Res Pract. 2022;2022:4549134. https://doi.org/10.1155/2022/4549134.
25. Lu X. A randomized, double-blind, controlled study of probiotics for the treatment of unusual psoriasis. Chin Contemp Med. 2017;24(08):47–9.
26. Harden JL, Krueger JG, Bowcock AM. The immunogenetics of psoriasis: a comprehensive review. J Autoimmun. 2015;64:66–73. https://doi.org/10.1016/j.jaut.2015.07.008.
27. Navarro-López V, Martínez-Andrés A, Ramírez-Boscà A, Ruzafa-Costas B, Núñez-Delegido E, Carrión-Gutiérrez MA, et al. Efficacy and safety of oral administration of a mixture of probiotic strains in patients with psoriasis: A randomized controlled clinical trial. Acta Dermato-Venereologica. 2019;99(12):1078–84. https://doi.org/10.2340/00015555-3305.
28. Mattei PL, Corey KC, Kimball AB. Psoriasis area severity index (PASI) and the dermatology life quality index (DLQI): the correlation between disease severity and psychological burden in patients treated with biological therapies. J Eur Acad Dermatol Venereol. 2014;28(3):333–7. https://doi.org/10.1111/jdv.12106.
29. Levkovich T, Poutahidis T, Smillie C, Varian BJ, Ibrahim YM, Lakritz JR, et al. Probiotic bacteria induce a 'glow of health.' PLoS One. 2013;8(1):e53867. https://doi.org/10.1371/journal.pone.0053867.
30. Akhter S, Tasnim FM, Islam MN, Rauf A, Mitra S, Emran TB, et al. Role of Th17 and IL-17 cytokines on inflammatory and auto-immune diseases. Curr Pharm Des. 2023;29(26):2078–90. https://doi.org/10.2174/1381612829666230904150808.
31. Pachauri A, Sharma S. Unravelling the gut-skin axis: the role of gut microbiota in pathogenesis and management of psoriasis. Inflammopharmacology. 2025. https://doi.org/10.1007/s10787-025-01813-y.
32. Zhu Q, Wu K, Yang Q, Meng B, Niu Y, Zhao F. Advances in psoriasis and gut microorganisms with co-metabolites. Front Microbiol. 2023;14: 1192543. https://doi.org/10.3389/fmicb.2023.1192543.
33. Chen M, Wang R, Wang T. Gut microbiota and skin pathologies: mechanism of the gut-skin axis in atopic dermatitis and psoriasis. Int Immunopharmacol. 2024;141: 112658. https://doi.org/10.1016/j.intimp.2024.112658.
34. Memariani M, Memariani H. New horizons in the treatment of psoriasis: modulation of gut microbiome. Heliyon. 2025;11(1):e41672. https://doi.org/10.1016/j.heliyon.2025.e41672.
35. Gottlieb AB, Langley RG, Philipp S, Sigurgeirsson B, Blauvelt A, Martin R, et al. Secukinumab improves physical function in subjects with plaque psoriasis and psoriatic arthritis: results from two randomized, phase 3 trials. J Drugs Dermatol. 2015;14(8):821–33.
36. Korneev A, Peshkova M, Koteneva P, Gundogdu A, Timashev P. Modulation of the skin and gut microbiome by psoriasis treatment: a comprehensive systematic review. Arch Dermatol Res. 2024;316(7):374. https://doi.org/10.1007/s00403-024-03024-x.
37. Bibbò S, Ianiro G, Gasbarrini A, Cammarota G. Fecal microbiota transplantation: past, present and future perspectives. Minerva Gastroenterol Dietol. 2017;63(4):420–30. https://doi.org/10.23736/s1121-421x.17.02374-1.
38. Zmora N, Zilberman-Schapira G, Suez J, Mor U, Dori-Bachash M, Bashiardes S, et al. Personalized gut mucosal colonization resistance to empiric probiotics is associated with unique host and Microbiome features. Cell. 2018;174(6):1388–e40521. https://doi.org/10.1016/j.cell.2018.08.041.
39. Mazziotta C, Tognon M, Martini F, Torreggiani E, Rotondo JC. Probiotics mechanism of action on immune cells and beneficial effects on human health. Cells. 2023;12(1): 184. https://doi.org/10.3390/cells12010184.
40. van Nood E, Vrieze A, Nieuwdorp M, Fuentes S, Zoetendal EG, de Vos WM, et al. Duodenal infusion of donor feces for recurrent *clostridium difficile*. N Engl J Med. 2013;368(5):407–15. https://doi.org/10.1056/NEJMoa1205037.
41. Owczarek W. The role of HLA-Cw6 in psoriasis and psoriatic arthritis. Reumatologia. 2022;60(5):303–5. https://doi.org/10.5114/reum.2022.120752.
42. Belkaid Y, Hand TW. Role of the microbiota in immunity and inflammation. Cell. 2014;157(1):121–. https://doi.org/10.1016/j.cell.2014.03.011. 41.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.